🟩 30 EJERCICIOS PRÁCTICOS DEL MÓDULO 6

✅ BÁSICOS (1–10)

1. Mostrar interfaces de red

- ip a

2. Mostrar la ruta por defecto

- ip r

3. Hacer ping a tu puerta de enlace

Ejemplo:

- ping 192.168.1.1

4. Consultar DNS con nslookup

- nslookup google.com

```
jcsantos@JCSantos-Ubunut:/etc/syste
/etc/systemd/system

┌jcsantos at JCSantos-Ubunut in /etc/systemd/system
└○ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:2c:60:2e brd ff:ff:ff:ff:ff:ff
    inet 192.168.2.132/24 brd 192.168.2.255 scope global dynamic noprefixroute ens33
       valid_lft 1676sec preferred_lft 1676sec
    inet6 fe80::20c:29ff:fe2c:602e/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
┌jcsantos at JCSantos-Ubunut in /etc/systemd/system
└○ ip r
default via 192.168.2.2 dev ens33 proto dhcp src 192.168.2.132 metric 100
192.168.2.0/24 dev ens33 proto kernel scope link src 192.168.2.132 metric 100
┌jcsantos at JCSantos-Ubunut in /etc/systemd/system
└○ ping 192.168.2.2
PING 192.168.2.2 (192.168.2.2) 56(84) bytes of data.
64 bytes from 192.168.2.2: icmp_seq=1 ttl=128 time=0.315 ms
64 bytes from 192.168.2.2: icmp_seq=2 ttl=128 time=0.618 ms
64 bytes from 192.168.2.2: icmp_seq=3 ttl=128 time=0.687 ms
64 bytes from 192.168.2.2: icmp_seq=4 ttl=128 time=0.611 ms
64 bytes from 192.168.2.2: icmp_seq=5 ttl=128 time=0.661 ms
64 bytes from 192.168.2.2: icmp_seq=6 ttl=128 time=0.618 ms
^C
--- 192.168.2.2 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5125ms
rtt min/avg/max/mdev = 0.315/0.585/0.687/0.123 ms
┌jcsantos at JCSantos-Ubunut in /etc/systemd/system
└○ nslookup google.com
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
Name:   google.com
Address: 142.251.142.142
Name:   google.com
Address: 2a00:1450:4003:80a::200e
```

5. Ver conexiones activas

- ss -tunap

```
jcsantos@JCSantos-Ubunut:/etc/systemd/system
/etc/systemd/system

jcsantos at JCSantos-Ubunut in /etc/systemd/system
ss -tunap
Netid State   Recv-Q  Send-Q        Local Address:Port      Peer Address:Port  Process
udp   UNCONN  0       0                127.0.0.54:53             0.0.0.0:*
udp   UNCONN  0       0             127.0.0.53%lo:53             0.0.0.0:*
udp   ESTAB   0       0      192.168.2.132%ens33:68        192.168.2.254:67
udp   UNCONN  0       0                 127.0.0.1:323            0.0.0.0:*
udp   UNCONN  0       0                  0.0.0.0:5353            0.0.0.0:*
udp   UNCONN  0       0                  0.0.0.0:55190           0.0.0.0:*
udp   UNCONN  0       0                    [::1]:323               [::]:*
udp   UNCONN  0       0                     [::]:5353              [::]:*
udp   UNCONN  0       0                     [::]:55802             [::]:*
tcp   LISTEN  0       4096             127.0.0.54:53             0.0.0.0:*
tcp   LISTEN  0       4096              127.0.0.1:631            0.0.0.0:*
tcp   LISTEN  0       80                127.0.0.1:3306           0.0.0.0:*
tcp   LISTEN  0       4096          127.0.0.53%lo:53             0.0.0.0:*
tcp   LISTEN  0       4096                0.0.0.0:22             0.0.0.0:*
tcp   LISTEN  0       4096                  [::1]:631              [::]:*
tcp   LISTEN  0       4096                   [::]:22               [::]:*
tcp   LISTEN  0       511                       *:80                  *:*
jcsantos at JCSantos-Ubunut in /etc/systemd/system
```

6. Usar traceroute

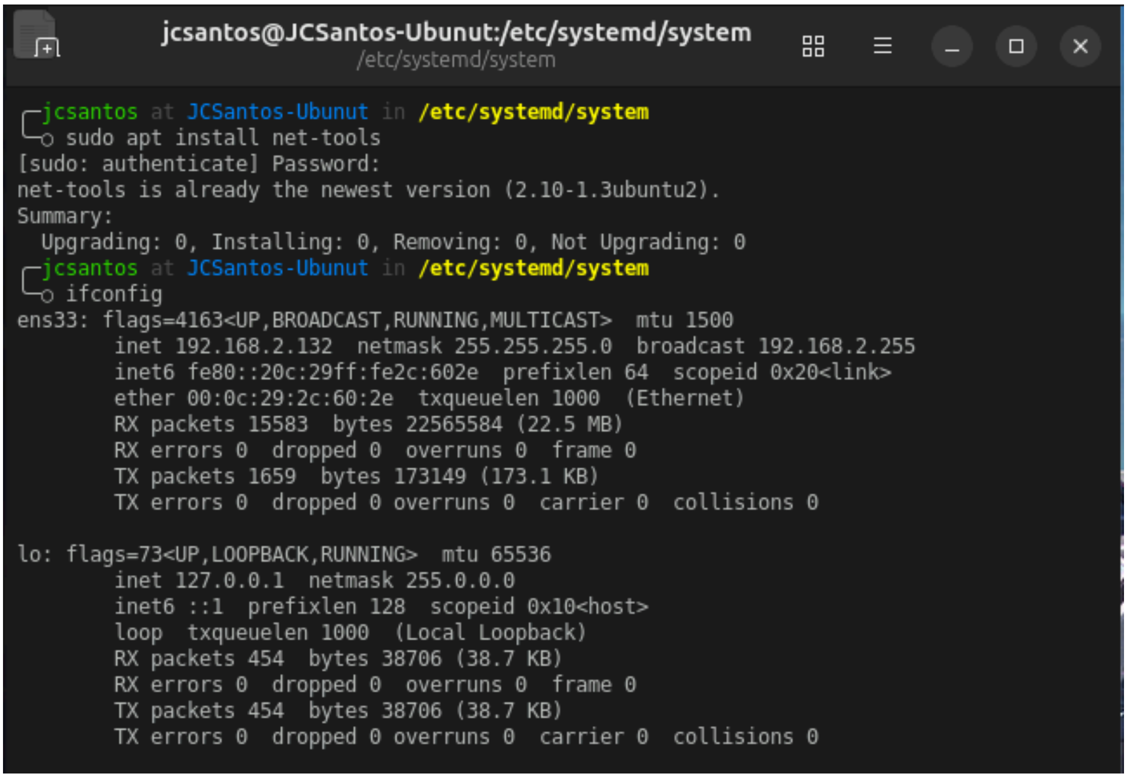
- traceroute 8.8.8.8

```
jcsantos@JCSantos-Ubunut:/etc/systemd/system
/etc/systemd/system

jcsantos at JCSantos-Ubunut in /etc/systemd/system
traceroute 8.8.8.8
traceroute to 8.8.8.8 (8.8.8.8), 30 hops max, 60 byte packets
 1  _gateway (192.168.2.2)  0.483 ms  0.411 ms  0.378 ms
 2  192.168.0.1 (192.168.0.1)  11.469 ms  11.439 ms  13.053 ms
 3  * * *
 4  * * *
 5  33.red-81-45-103.staticip.rima-tde.net (81.45.103.33)  26.100 ms  26.070 ms  26.042 ms
 6  * * *
 7  94.red-81-46-8.customer.static.ccgg.telefonica.net (81.46.8.94)  18.330 ms  19.595 ms  19.522 ms
 8  176.52.253.93 (176.52.253.93)  19.468 ms  19.429 ms  19.324 ms
 9  5.53.0.176 (5.53.0.176)  19.262 ms google-be4-grcmadno1.net.telefonicaglobalsolutions.com (213.140.50.43)  19.103 ms 72.14.219.20 (72.14.219.20)  20.128 ms
10  192.178.110.71 (192.178.110.71)  20.037 ms  19.993 ms 108.170.252.213 (108.170.252.213)  19.949 ms
11  74.125.253.197 (74.125.253.197)  20.370 ms 142.251.49.55 (142.251.49.55)  20.327 ms 142.251.51.141 (142.251.51.141)  20.290 ms
12  dns.google (8.8.8.8)  18.087 ms  17.987 ms  21.469 ms
jcsantos at JCSantos-Ubunut in /etc/systemd/system
```

7. Instalar net-tools y usar ifconfig *(opcional)*

- sudo apt install net-tools
- ifconfig



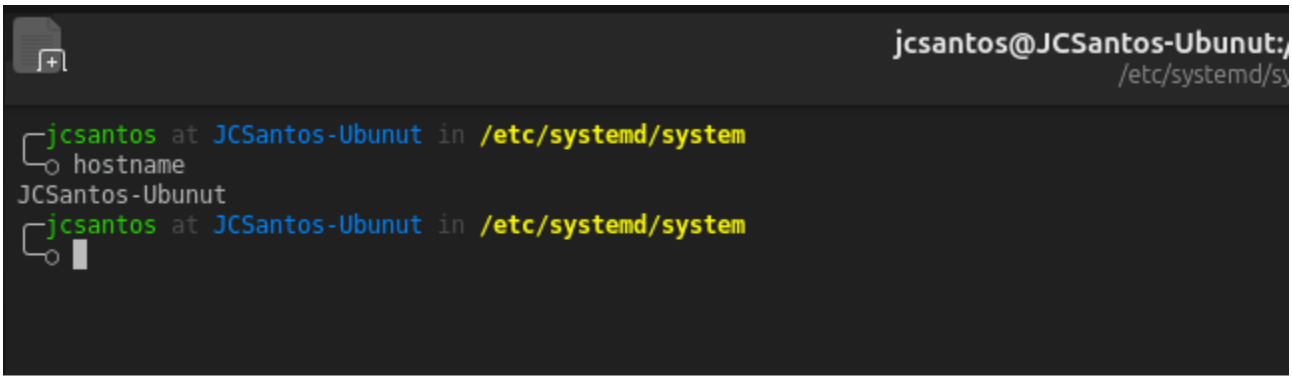
```
jcsantos@JCSantos-Ubunut:/etc/systemd/system
/etc/systemd/system

jcsantos at JCSantos-Ubunut in /etc/systemd/system
  sudo apt install net-tools
[sudo: authenticate] Password:
net-tools is already the newest version (2.10-1.3ubuntu2).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0
jcsantos at JCSantos-Ubunut in /etc/systemd/system
  ifconfig
ens33: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.2.132  netmask 255.255.255.0  broadcast 192.168.2.255
        inet6 fe80::20c:29ff:fe2c:602e  prefixlen 64  scopeid 0x20<link>
        ether 00:0c:29:2c:60:2e  txqueuelen 1000  (Ethernet)
        RX packets 15583  bytes 22565584 (22.5 MB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 1659  bytes 173149 (173.1 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 454  bytes 38706 (38.7 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 454  bytes 38706 (38.7 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

8. Ver tu hostname

- hostname



```
jcsantos@JCSantos-Ubunut:/
/etc/systemd/sy

jcsantos at JCSantos-Ubunut in /etc/systemd/system
  hostname
JCSantos-Ubunut
jcsantos at JCSantos-Ubunut in /etc/systemd/system
```

9. Modificar temporalmente tu IP

- sudo ip a add 192.168.1.200/24 dev enp0s3

```
┌jcsantos at JCSantos-Ubunut in /etc/systemd/system
└o ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:2c:60:2e brd ff:ff:ff:ff:ff:ff
    inet 192.168.2.132/24 brd 192.168.2.255 scope global dynamic noprefixroute ens33
       valid_lft 1138sec preferred_lft 1138sec
    inet6 fe80::20c:29ff:fe2c:602e/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
┌jcsantos at JCSantos-Ubunut in /etc/systemd/system
└o ping 192.168.2.232
PING 192.168.2.232 (192.168.2.232) 56(84) bytes of data.
From 192.168.2.132 icmp_seq=1 Destination Host Unreachable
From 192.168.2.132 icmp_seq=2 Destination Host Unreachable
From 192.168.2.132 icmp_seq=3 Destination Host Unreachable
^C
--- 192.168.2.232 ping statistics ---
5 packets transmitted, 0 received, +3 errors, 100% packet loss, time 4100ms
pipe 3
┌jcsantos at JCSantos-Ubunut in /etc/systemd/system
└o sudo ip a add 192.168.2.232/24 dev ens33
┌jcsantos at JCSantos-Ubunut in /etc/systemd/system
└o ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:2c:60:2e brd ff:ff:ff:ff:ff:ff
    inet 192.168.2.132/24 brd 192.168.2.255 scope global dynamic noprefixroute ens33
       valid_lft 978sec preferred_lft 978sec
    inet 192.168.2.232/24 scope global secondary ens33
       valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fe2c:602e/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
┌jcsantos at JCSantos-Ubunut in /etc/systemd/system
└o
```

10. Quitar una IP añadida

- sudo ip a del 192.168.1.200/24 dev enp0s3

11. Repite el ejercicio anterior con otra ip

```
jcsantos@JCSantos-Ubunut:/etc/systemd/syst
/etc/systemd/system

┌jcsantos at JCSantos-Ubunut in /etc/systemd/system
└o sudo ip a del 192.168.2.232/24 dev ens33
┌jcsantos at JCSantos-Ubunut in /etc/systemd/system
└o ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:2c:60:2e brd ff:ff:ff:ff:ff:ff
    inet 192.168.2.132/24 brd 192.168.2.255 scope global dynamic noprefixroute ens33
       valid_lft 1791sec preferred_lft 1791sec
    inet6 fe80::20c:29ff:fe2c:602e/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
┌jcsantos at JCSantos-Ubunut in /etc/systemd/system
└o
```

✅ INTERMEDIOS (12–20)

12. Aplicar la configuración

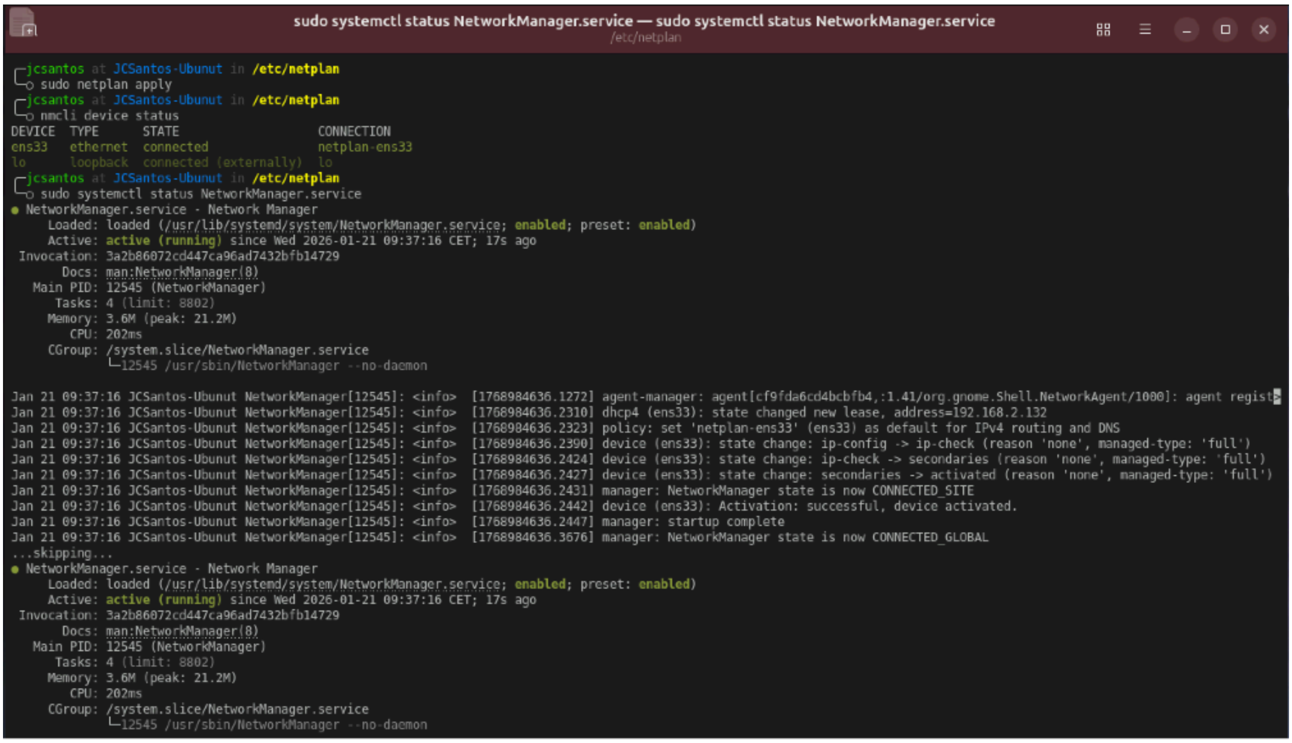
- sudo netplan apply

13. Ver estado de NetworkManager (Desktop)

- nmcli device status

14. Revisar logs del servicio de red

- journalctl -u NetworkManager



---

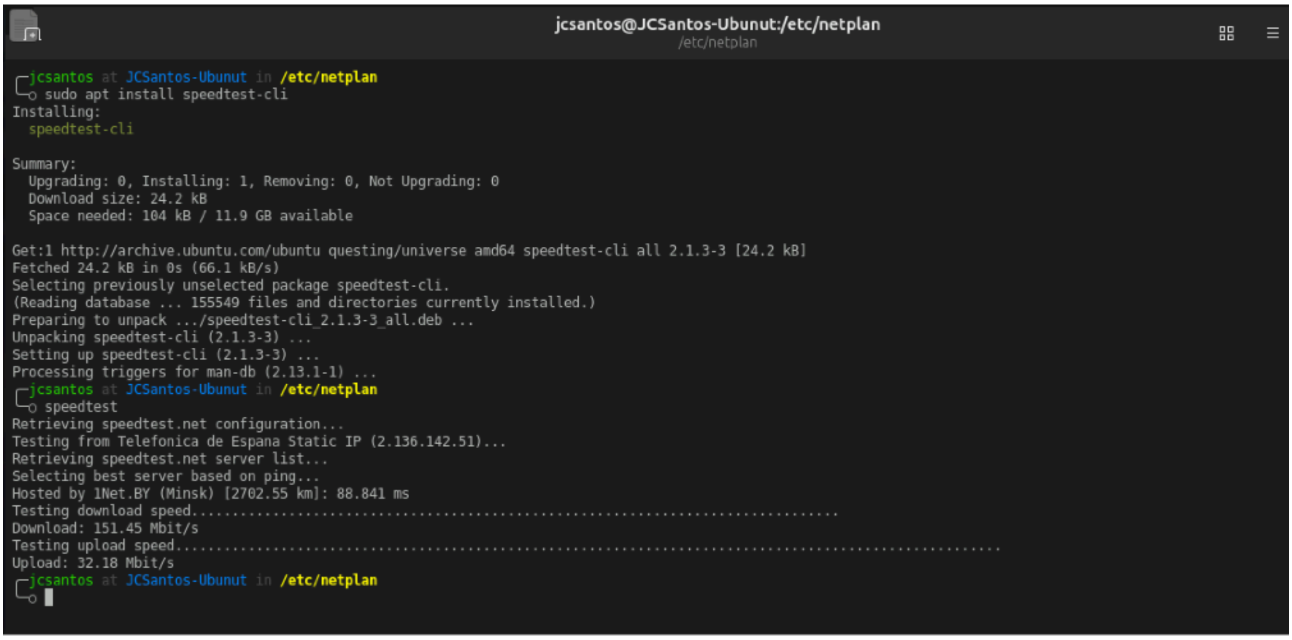
15. Escanear redes WiFi disponibles

- nmcli device wifi list

Al estar en VM no tengo el adaptador WiFi configurado.

16. Hacer un test de velocidad simple

- sudo apt install speedtest-cli
- speedtest



```
jcsantos@JCSantos-Ubunut:/etc/netplan
/etc/netplan

jcsantos at JCSantos-Ubunut in /etc/netplan
  sudo apt install speedtest-cli
Installing:
  speedtest-cli

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 0
  Download size: 24.2 kB
  Space needed: 104 kB / 11.9 GB available

Get:1 http://archive.ubuntu.com/ubuntu questing/universe amd64 speedtest-cli all 2.1.3-3 [24.2 kB]
Fetched 24.2 kB in 0s (66.1 kB/s)
Selecting previously unselected package speedtest-cli.
(Reading database ... 155549 files and directories currently installed.)
Preparing to unpack .../speedtest-cli_2.1.3-3_all.deb ...
Unpacking speedtest-cli (2.1.3-3) ...
Setting up speedtest-cli (2.1.3-3) ...
Processing triggers for man-db (2.13.1-1) ...
jcsantos at JCSantos-Ubunut in /etc/netplan
  speedtest
Retrieving speedtest.net configuration...
Testing from Telefonica de Espana Static IP (2.136.142.51)...
Retrieving speedtest.net server list...
Selecting best server based on ping...
Hosted by 1Net.BY (Minsk) [2702.55 km]: 88.841 ms
Testing download speed................................................................................
Download: 151.45 Mbit/s
Testing upload speed....................................................................................................
Upload: 32.18 Mbit/s
jcsantos at JCSantos-Ubunut in /etc/netplan
```

17. Abrir puerto para un servicio

- sudo ufw allow 8080/tcp

18. Bloquear puerto

- sudo ufw deny 23/tcp

19. Ver reglas activas de firewall

- sudo ufw status verbose

```
jcsantos@JCSantos-Ubunut:/etc/netplan
/etc/netplan

┌jcsantos at JCSantos-Ubunut in /etc/netplan
└○ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip
┌jcsantos at JCSantos-Ubunut in /etc/netplan
└○ sudo ufw allow 8080/tcp
Rule added
Rule added (v6)
┌jcsantos at JCSantos-Ubunut in /etc/netplan
└○ sudo ufw deny 23/tcp
Rule added
Rule added (v6)
┌jcsantos at JCSantos-Ubunut in /etc/netplan
└○ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

To                         Action      From
--                         ------      ----
8080/tcp                   ALLOW IN    Anywhere
23/tcp                     DENY IN     Anywhere
8080/tcp (v6)              ALLOW IN    Anywhere (v6)
23/tcp (v6)                DENY IN     Anywhere (v6)

┌jcsantos at JCSantos-Ubunut in /etc/netplan
└○
```

20. Conectarte por SSH desde otra máquina

- ssh usuario@IP

```
┌jcsantos at JCSantos-Ubunut in /etc/netplan
└○ ssh jcsantos@servidor.santosarias.es -p 2024
The authenticity of host '[servidor.santosarias.es]:2024 ([217.160.248.16]:2024)' can't be established.
ED25519 key fingerprint is SHA256:oMEPQJyYCR29XN/JV+eMy1udxqkk/5BG34FhAVhrza0.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[servidor.santosarias.es]:2024' (ED25519) to the list of known hosts.
jcsantos@servidor.santosarias.es's password:
Linux VPN-Unifi 6.1.0-37-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.140-1 (2025-05-22) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Fri Jul  4 12:41:24 2025 from 176.87.26.114
jcsantos@VPN-Unifi:~$
```

✅ AVANZADOS (21–30)

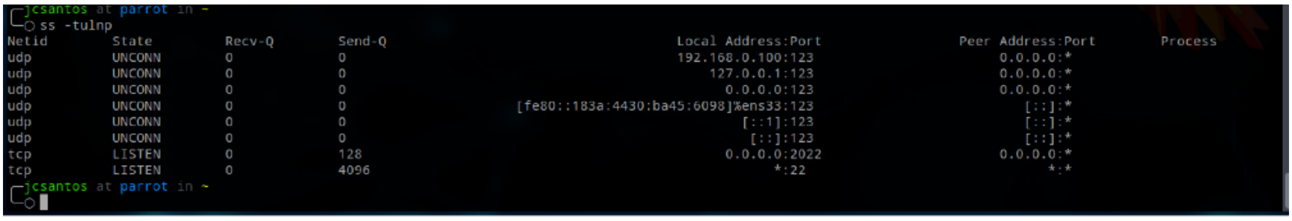
21. Cambiar el puerto por defecto de SSH

Editar:

- sudo nano /etc/ssh/sshd_config

22. Comprobar que SSH escucha en el nuevo puerto

- ss -tulnp l grep ssh

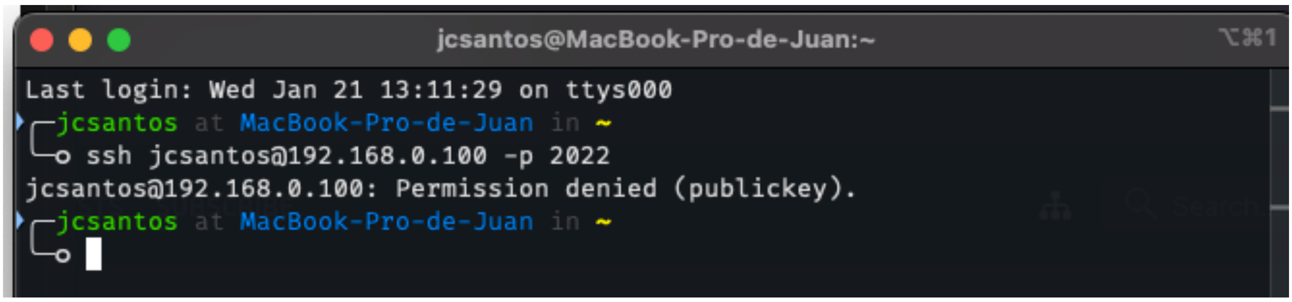


---

23. Forzar autenticación por clave pública

Modificar:

- PasswordAuthentication no



24. Probar fallo de DNS

Cambiar DNS a uno incorrecto y probar ping a dominio.

```
┌─jcsantos at parrot in ~
└─○ ping google.es
ping: google.es: Temporary failure in name resolution
┌─jcsantos at parrot in ~
└─○ 
```

---

25. Analizar tráfico activo por proceso

- ss -tp

```
┌─jcsantos at parrot in ~
└─○ ss -tp
State   Recv-Q   Send-Q    Local Address:Port      Peer Address:Port   Process
ESTAB   0        0         192.168.0.209:44216     34.107.221.82:http    users:(("firefox-esr",pid=2125,fd=139))
ESTAB   0        0         192.168.0.209:48414     2.16.54.183:https     users:(("firefox-esr",pid=2125,fd=163))
ESTAB   0        0         192.168.0.209:47260     151.101.1.91:https    users:(("firefox-esr",pid=2125,fd=15))
ESTAB   0        0         192.168.0.209:50794     52.142.124.215:https  users:(("firefox-esr",pid=2125,fd=172))
ESTAB   0        0         192.168.0.209:53762     34.160.144.191:https  users:(("firefox-esr",pid=2125,fd=70))
ESTAB   0        0         192.168.0.209:51288     74.178.116.247:https  users:(("firefox-esr",pid=2125,fd=174))
ESTAB   0        0         192.168.0.209:48380     2.16.54.183:https     users:(("firefox-esr",pid=2125,fd=154))
ESTAB   0        0         192.168.0.209:58422     34.107.243.93:https   users:(("firefox-esr",pid=2125,fd=120))
ESTAB   0        0         192.168.0.209:54650     142.250.200.131:http  users:(("firefox-esr",pid=2125,fd=93))
ESTAB   0        0         192.168.0.209:48610     142.251.208.227:http  users:(("firefox-esr",pid=2125,fd=118))
ESTAB   0        0         192.168.0.209:54666     142.250.200.131:http  users:(("firefox-esr",pid=2125,fd=136))
ESTAB   0        0         192.168.0.209:52846     52.142.125.222:https  users:(("firefox-esr",pid=2125,fd=178))
ESTAB   0        0         192.168.0.209:52016     23.227.38.74:https    users:(("firefox-esr",pid=2125,fd=160))
ESTAB   0        0         192.168.0.209:58430     34.107.243.93:https   users:(("firefox-esr",pid=2125,fd=128))
┌─jcsantos at parrot in ~
└─○
```

26. trabajo en equipo: Crear una red virtual interna entre varias VMs

- Asignar IPs manualmente y probar ping entre ellas.

27. Hacer port forwarding en VirtualBox

Acceder por SSH desde tu PC Host.

```
┌─jcsantos at MacBook-Pro-de-Juan in ~
└─○ ssh jcsantos@192.168.0.209 -p 2022
The authenticity of host '[192.168.0.209]:2022 ([192.168.0.209]:2022)' can't be established.
ED25519 key fingerprint is SHA256:SakbN4IYzWHi6gUSwcXmaAP1rXz6A+BZJpFn1Z557NY.
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:4: [192.168.0.100]:2022
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[192.168.0.209]:2022' (ED25519) to the list of known hosts.
jcsantos@192.168.0.209's password:
Linux parrot 6.12.32-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.32-1parrot1 (2025-06-27) x86_64


The programs included with the Parrot GNU/Linux are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Parrot GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Wed Jan 21 12:13:35 2026 from 192.168.0.123
┌─jcsantos at parrot in ~
└─○ 
```

28. Crear un script de diagnóstico de red

Debe mostrar:

- IP
- Gateway
- DNS
- Conectividad a Internet

```
-rwxr-xr-x 1 jcsantos jcsantos  837 Jan 21 13:02 info_Red.sh
┌─jcsantos at parrot in ~
└─○ nvim info_Red.sh
┌─jcsantos at parrot in ~
└─○ cat info_Red.sh
#!/bin/bash

echo "=== DIAGNÓSTICO DE RED ==="
echo "Fecha: $(date)"
echo ""

# IP
echo "IP:"
ip route show default | grep default | awk '{print $3 " (Gateway)"}' || echo "No IP detectada"
ip addr show | grep "inet " | awk '{print $2}' | head -5 || echo "Sin IP"
echo ""

# Gateway
echo "Gateway:"
ip route | grep default | awk '{print $3}' | head -1 || echo "No gateway"
echo ""

# DNS
echo "DNS:"
cat /etc/resolv.conf | grep nameserver | head -3 || echo "No DNS configurados"
echo ""

# Conectividad Internet
echo "Conectividad:"
ping -c 3 8.8.8.8 > /dev/null 2>&1
if [ $? -eq 0 ]; then
    echo "✓ Internet OK (Google DNS)"
    curl -s --max-time 10 http://www.google.com > /dev/null && echo "✓ Web OK" || echo "✗ Web falló"
else
    echo "✗ Sin Internet"
fi
┌─jcsantos at parrot in ~
└─○ █
```

29. Simular caída de red

- sudo ip link set enp0s3 down

Y recuperarla:

- sudo ip link set enp0s3 up

---

30. Capturar paquetes (Wireshark o tcpdump)

Ejemplo:

- sudo tcpdump -i enp0s3 -c 20

🛠️ MINI–PROYECTO DEL MÓDULO 6 EN EQUIPO

Objetivo:

Configurar **un servidor Ubuntu accesible por SSH de forma segura** y documentar todo el proceso.

Tareas:

1. Asignar **IP estática** usando Netplan.
2. Configurar **DNS manuales**.
3. Activar SSH y cambiar puerto.
4. Crear clave pública/privada y desactivar login por contraseña.
5. Configurar firewall UFW:
    - permitir nuevo puerto SSH
    - denegar puertos innecesarios
6. Crear script diagnostico_red.sh con:
    - IP actual
    - Gateway
    - DNS
    - Ping a 8.8.8.8
    - Ping a google.com
7. Probar conexión remota desde otra máquina.
8. Documentar todo:

- Archivos modificados
- Comandos usados
- Problemas encontrados
- Solución aplicada